

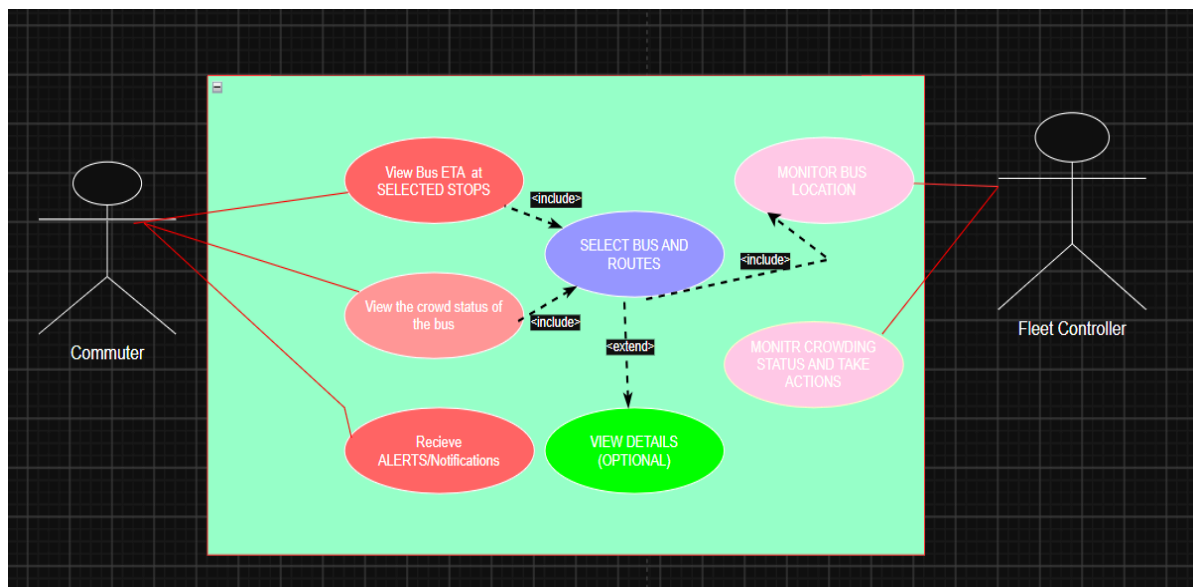
NAME:SAIRAM CHARAN NUKANA
CLASS/SEC:5-E
SRN:PES1UG24AM240

USE CASE NO:24

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall collect bus GPS coordinates every 5 seconds and update the estimated arrival times for upcoming stops.	High	ETA is recalculated using the latest GPS information and changes appropriately as the bus moves.	Helps commuters know when the bus will arrive.
FR-002	Functional	The system shall allow commuters to view the current crowding status of a bus.	High	When a commuter selects a bus, its current crowding level is displayed.	Helps commuters decide whether to take the bus.
FR-003	Functional	The system shall allow the Fleet Controller to monitor the current location and crowding status of buses.	High	When the Fleet Controller selects a bus, its GPS location and crowding status are displayed.	Helps the controller monitor bus operations.
FR-004	Functional	The system shall allow commuters to view the estimated arrival time of a bus at their selected stop.	High	When the commuter selects a bus and stop, the system displays the estimated arrival time.	Helps commuters plan their journey and reduce waiting time.
FR-005	Functional	The system shall display the computed passenger crowding level for each bus	High	When updated ticketing data is received, the system computes and updates the	Keeps crowding information up to date.

		based on ticketing sensor data.		bus's crowding status.	
NFR-001	Nonfunctional	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Benchmark testing confirms that the system handles 1,000 active vehicles under the required performance conditions.	Ensures the system can support a large city-wide fleet.
NFR-002	Nonfunctional	The system shall load and display the latest bus tracking information within 5 seconds of a commuter's request.	High	When the commuter requests the latest information, the system displays the updated location, ETA, and crowding information within 5 seconds.	Ensures commuters receive timely live information .

USE CASE DIAGRAM



Use-Case Flow – Track Bus and View ETA

Preconditions

- The bus routes should be available in the system.
- The buses should be sending their GPS location.

Main Success Scenario

1. The commuter searches for the bus route they want.
2. The commuter selects the bus.
3. The system shows where the bus is currently located.
4. The system shows how crowded the bus is.
5. The system shows the estimated time for the bus to reach the selected stop.
6. The commuter looks at this information and decides whether they want to take the bus.

Postcondition

The commuter gets the bus location, ETA, and crowding information and can decide whether to take the bus.

Alternate Flow – GPS Data Not Available

1. The commuter selects a bus route.
2. The system is not able to get the GPS information of the bus.
3. The system shows a message saying, "Sorry! Unable to fetch bus tracking data."
4. The commuter can try again or select another bus.